Hints: Polynomials – 1. Roots, multiplicity and the derivative

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Work over $\mathbb{C}$. The divisibility $f'^q \mid f^p$ says that every root of f' is a root of f.

Hint 2. Let f have distinct roots $z_1, \dots, z_m$ with multiplicities $k_1, \dots, k_m$. Then z_i is a root of f' of multiplicity exactly $k_i - 1$, which accounts for $\sum(k_i - 1) = n - m$ of the $n - 1$ roots of f'.

Hint 3. The remaining $m - 1$ roots of f' are not roots of f, so by Hint 1 there are none: $m = 1$. Hence $f = c(x - a)^n$ and $f' = nc(x - a)^{n-1}$ divides f.

Problem 2

Hint 1. The statement is invariant under translations and positive dilations of x (the parts of (b, c) move along). Normalize $b = 0$, $c = 1$, so $a = -t$ with $t \geqslant 0$ and $f(x) = x(x - 1)(x + t)$.

Hint 2. Then $f'(x) = 3x^2 + 2(t - 1)x - t$. Since $f'(0) = -t \leqslant 0 < f'(1)$ and f' is quadratic with positive leading coefficient, f' has exactly one root in $(0, 1)$ (when $t = 0$ the other root is 0); locate it by evaluating f' at $\frac{1}{2}$ and $\frac{2}{3}$.

Hint 3. $f'(\frac{1}{2}) = -\frac{1}{4} < 0$ and $f'(\frac{2}{3}) = \frac{t}{3} \geqslant 0$, so the root lies in $(\frac{1}{2}, \frac{2}{3}]$, the fourth sixth. It equals $\frac{2}{3}$ exactly when $t = 0$, i.e. $a = b$ and $f = (x - b)^2(x - c)$; it never equals $\frac{1}{2}$ (it only tends to it as $a \to -\infty$).

Problem 3

Hint 1. Compare multiplicities: suppose r is a root of P of multiplicity $m \geqslant 2$. What is its multiplicity in P'', and hence in Q?

Hint 2. From $P = QP''$ you get $m = \mathrm{mult}_r(Q) + (m - 2)$, so $Q = c(x - r)^2$; comparing leading coefficients, $c = \frac{1}{n(n-1)}$.

Hint 3. Translate $r = 0$: $n(n - 1)P = x^2P''$. Compare the coefficients of x^k: $n(n - 1)a_k = k(k - 1)a_k$, so $a_k = 0$ for $k \neq n$ and $P = a_n x^n$ has only one root. So if P has two distinct roots, all roots are simple.

Problem 4

Hint 1. Use the logarithmic derivative $\dfrac{p'(x)}{p(x)} = \sum_{j=1}^{n} \dfrac{1}{x - r_j}$ (all roots are the r_j, simple).

Hint 2. At the midpoint $m_i = \frac{r_i + r_{i+1}}{2}$ the terms $j = i$ and $j = i + 1$ cancel. Look at $i = 1$ when $n \geqslant 3$: what is the sign of each of the remaining terms?

Hint 3. For $i = 1$ all remaining terms $\frac{1}{m_1 - r_j}$, $j \geqslant 3$, are negative, so $p'(m_1) \neq 0$ when $n \geqslant 3$. Answer: exactly the quadratics $p(x) = a(x - r_1)(x - r_2)$ with $r_1 < r_2$, $a \neq 0$ ($n = 2$).

Problem 5

Hint 1. Factor p over $\mathbb{C}$. Why must every real root have even multiplicity, and why is the leading coefficient positive?

Hint 2. The non-real roots come in conjugate pairs. Put $g(x) = \prod(x - z)$, taking one root from each conjugate pair and half of each real root's multiplicity. Then $p = c\, g\, \bar{g}$ with $c > 0$.

Hint 3. Write $g = A + iB$ with A, B real polynomials; then $p = c(A^2 + B^2) = (\sqrt{c}A)^2 + (\sqrt{c}B)^2$, so $k = 2$ suffices.